

MUYIDEEN ABDULSAMAD

Research Engineer · AI-Driven Materials Discovery · MLOps · Scientific Workflow Automation

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PROFESSIONAL SUMMARY

Materials engineer and ML software developer with 7+ years building production AI systems and 4+ years of hands-on computational materials research. Uniquely positioned for autonomous materials lab infrastructure: holds a B.Eng. in Metallurgical & Materials Engineering combined with production-grade expertise in Python, Kubernetes, Docker, MongoDB, and scientific ML pipeline development. Built a CFD surrogate model (digital twin replacing COMSOL solver at ~10x speedup), a materials inverse design system, and a composition-to-property prediction pipeline — all deployed on cloud-native infrastructure. Conversational Arabic. Awarded the 2024 Computational Materials Engineer Class Award for innovation in ML-driven materials research.

TECHNICAL SKILLS

Python & Scientific Computing: Python (advanced, 7 years), NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, SciPy, Matplotlib — used daily for materials ML and data pipelines

MLOps & Cloud-Native Infra: Kubernetes, Docker, CI/CD, FastAPI, REST APIs, MLflow, Git, Linux/Bash (advanced), Nginx — production deployments serving 100+ users

Data Engineering & Scientific Data Platforms: MongoDB (intermediate), JSON, materials dataset curation, high-throughput data pipeline design, ANSYS Granta, Materials Project API

Digital Twins & Simulation Surrogates: CFD surrogate modelling (COMSOL → neural network), ANSYS FEA, COMSOL Multiphysics, physics-informed neural networks, simulation workflow automation

Scientific Workflow Orchestration: End-to-end ML pipeline design: data ingestion → feature engineering → model training → validation → containerised deployment → monitoring

Materials Science Domain: Inorganic materials, solid-state physics, thermodynamics, phase diagram analysis, SEM/TEM/XRD characterisation, NDT, composite fabrication, powder metallurgy, sol-gel synthesis

Autonomous & Reinforcement Learning: Materials inverse design (goal-directed generation), Bayesian optimisation concepts, active learning frameworks — developing RL for experiment planning

Languages & Communication: Python, Rust, Java, JavaScript, SQL, MATLAB, Bash — English (proficient), Arabic (conversational), Yoruba/Hausa (fluent)

RELEVANT EXPERIENCE

Digital Twin & Surrogate Model Developer | Independent Research

2024

- ▶ Built a physics-simulation surrogate neural network replacing COMSOL Multiphysics CFD solver runs — a working digital twin achieving ~10x simulation speedup while maintaining engineering-grade accuracy; directly applicable to the initiative's digital twin and autonomous experimentation goals.
- ▶ Surrogate architecture: feed-forward network trained on geometry-boundary condition pairs mapped to steady-state outputs; demonstrated that ML can replace expensive physics runs in iterative materials design workflows.
- ▶ Developed automated data collection pipeline from COMSOL simulation outputs; cleaned, normalised, and structured simulation results into training datasets full scientific data platform workflow.

Materials Informatics & Inverse Design Researcher | Independent / ABU Zaria

2022 – 2024

- ▶ Built a materials inverse design model given a target set of mechanical properties, predicts the elemental composition and heat treatment route required to achieve them; this is the core of autonomous experiment planning: AI proposes the next experiment based on desired outcome.
- ▶ Developed an end-to-end ML pipeline predicting mechanical properties of steel from iron-carbon phase diagram data 90%+ accuracy on held-out test data; full pipeline from raw phase diagram data through feature engineering, model training, validation, and prediction output.

- ▶ Used ANSYS Granta to systematically screen 15+ candidate materials and built structured datasets linking composition, processing route, and performance experience designing scientific data platforms for materials discovery workflows.
- ▶ Undergraduate research projects: bioglass synthesis (sol-gel), PET plastic → 3D printing filament recycling, aluminium-glass powder composites via powder metallurgy broad experimental materials domain knowledge underpinning the computational work.

ML Engineer — Production AI Systems | [Freelance / DataVessel Lab / STEM Coders](#)

2022 – Present

- ▶ Architected and deployed an enterprise management system (FastAPI + React.js) on Kubernetes with automated ML-powered data pipelines production cloud-native research infrastructure serving 100+ users; demonstrates the platform engineering capability the initiative requires.
- ▶ Built OPsonic for DataVessel Lab: designed full ML pipeline (data → features → CNN+LSTM model → REST microservice deployment); 92% classification accuracy, containerised on Docker and deployed as a production API representative of the scientific workflow orchestration role requires.
- ▶ Built and deployed all ML microservices using Docker and Kubernetes; implemented monitoring, CI/CD, and REST endpoints for model inference MLOps lifecycle from experiment to production.
- ▶ Developed FajrGuard custom CNN trained from scratch (PyTorch), quantised, and deployed on mobile for under 100ms on-device inference; demonstrates rapid prototyping from problem definition to working system.

Materials Analysis & CAE Engineer | [431 Engineering, Nigerian Air Force / UC San Diego Collaboration](#)

2022 – 2024

- ▶ Collaborated with UC San Diego Bioengineering doctoral researcher on bio-neuron chip design designed micro-electrode arrays and microfluidic channels informed by nanoscale fabrication constraints; experience integrating domain science with engineering design, the precise combination the autonomous lab initiative requires.
- ▶ Executed NDT inspections and materials characterisation (SEM, TEM, XRD) on aerospace composite components the experimental materials science foundation that makes autonomous lab design decisions meaningful rather than purely computational.
- ▶ Performed ANSYS Explicit Dynamics simulation of SiC-lead composite armour; characterised stress wave propagation across parameter sweeps running and analysing systematic simulation studies, the computational side of the closed-loop discovery workflow.

Engineering Designer — Defence (NDA) | [Technoguard / DICON \(Nigerian Defence Industries Corporation\)](#)

2023 – 2024

- ▶ *Engaged under NDA — specific deliverables are confidential.*
- ▶ Delivered engineering design software and tooling for a defence-sector engagement; applied systems-level design methodology and operated within a high-accountability, regulated project environment.

EDUCATION

B.Eng. Metallurgical & Materials Engineering

Oct 2024

Ahmadu Bello University, Zaria · Second Class Upper (Hons)

Thesis: ML simulation and prediction of steel mechanical properties via iron-carbon phase diagram — 90%+ prediction accuracy.

AWARDS & CERTIFICATIONS

- ▶ Computational Materials Engineer Class Award — ABU MME Department, 2024 (ML-driven materials innovation)
- ▶ Tech Oracle Award — NUESA Online Quiz Competition, 2024
- ▶ Google Project Management Certificate · Practical Ethical Hacking — TCM Academy
- ▶ Certified Network & Web Pentesting Professional — XSS-RAT

LANGUAGES

- ▶ English (Proficient) · Arabic (Conversational) · Hausa (Fluent) · Yoruba (Fluent) · French (Basic)